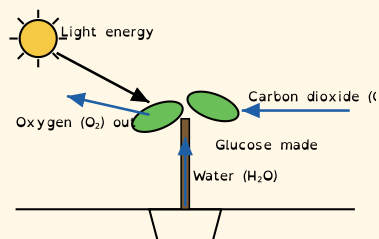


What is photosynthesis?

Photosynthesis is how green plants make their own food. They use light energy to turn carbon dioxide and water into glucose (a sugar) and oxygen. It happens in the leaves, inside tiny parts of the cell called chloroplasts.



Inputs and outputs of photosynthesis

Word & symbol equation

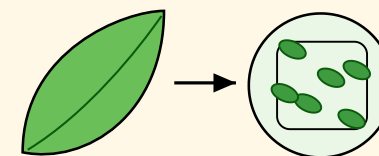
carbon dioxide + water $\rightarrow$ (light + chlorophyll) $\rightarrow$ glucose + oxygen

**Common mistakes**

- Mixing up the gases. A plant takes IN carbon dioxide and gives OUT oxygen — not the other way round.
- Calling light a 'raw material'. The raw materials are water and carbon dioxide; light is the energy.
- Saying it happens 'in the leaf'. Be exact: it happens in the chloroplast inside the cell.
- Forgetting plants respire too. Plants photosynthesise AND respire (respiration never stops).

Key vocabulary

- Photosynthesis** — how a plant makes its own food using light
- Chlorophyll** — the green colour that traps light energy
- Chloroplast** — the part of the cell where it happens
- Glucose** — the sugar (food) the plant makes
- Stomata** — tiny holes under the leaf for gases

In the leaf

Chloroplasts (green) in

Light is trapped by chlorophyll in the chloroplast

Questions — write your answer on the line

1. Which gas does a plant take IN for photosynthesis?

Answer: _____

2. Which gas does a plant give OUT?

Answer: _____

3. Name the part of the cell where photosynthesis happens.

Answer: _____

4. Complete: carbon dioxide + _____ $\rightarrow$ glucose + _____

Answer: _____